

인공지능 하드웨어 (2024-2학기)

Practice 2. Matrix Tiling

권 은 지 (EUNJI KWON)

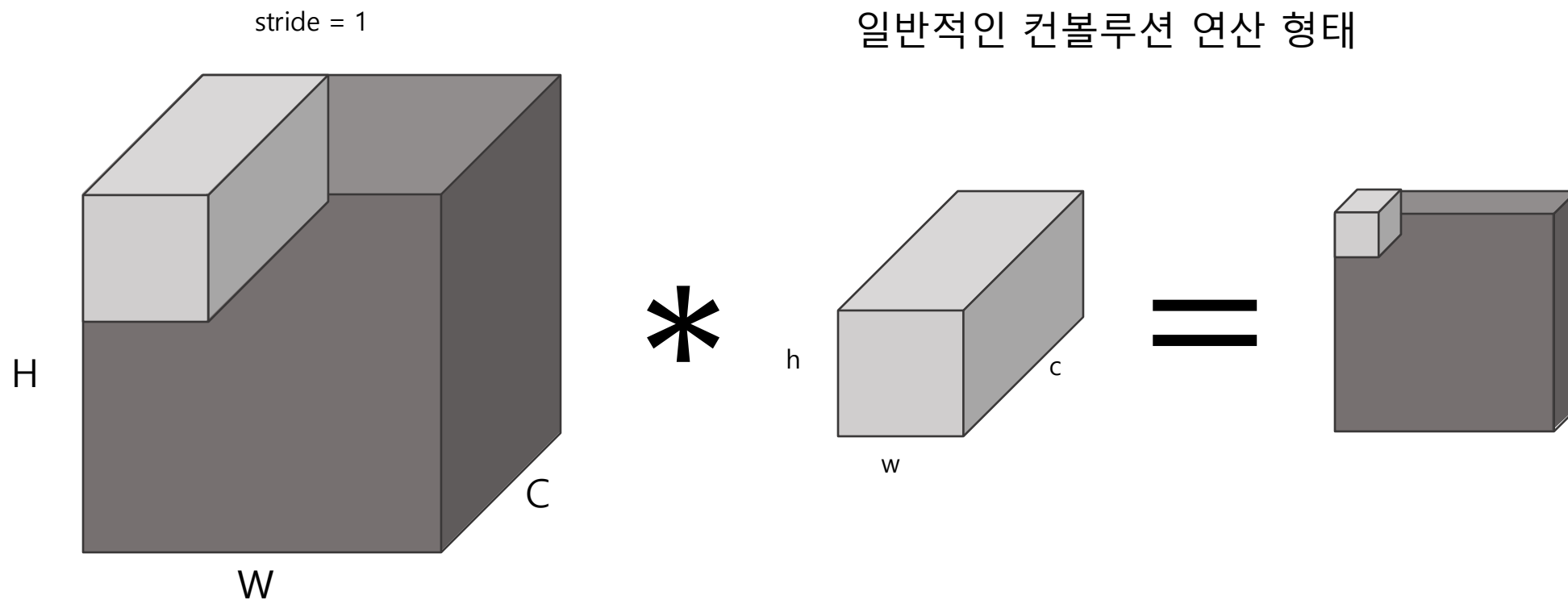
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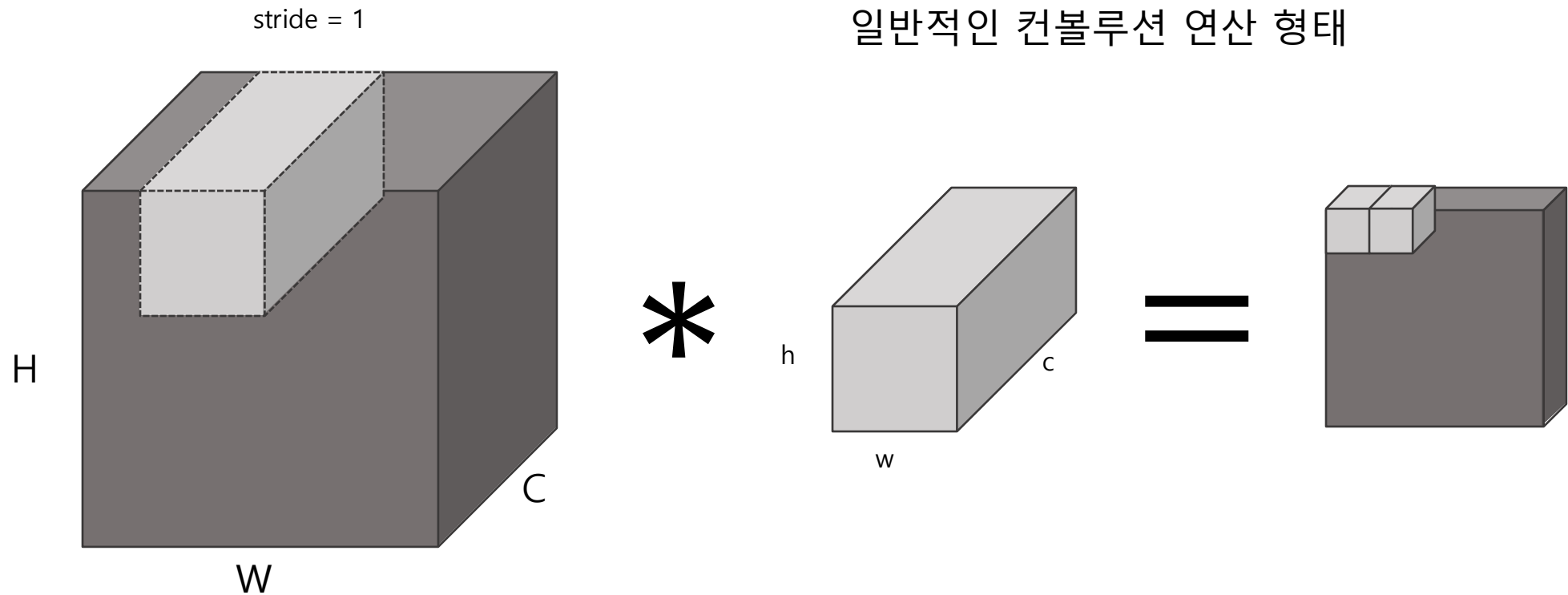
Goal

- **Implement optimized code for DNN forward computation**
- **Code Execution time 비교**
 - 1) Convolution (Practice1)
 - 2) Convolution with Matrix Tiling (implement your code with minimal library)
 - Tile size = 4
 - Tile size = 64
 - 3) Convolution with Matrix Tiling + np.matmul 사용
 - Tile size = 4
 - Tile size = 64

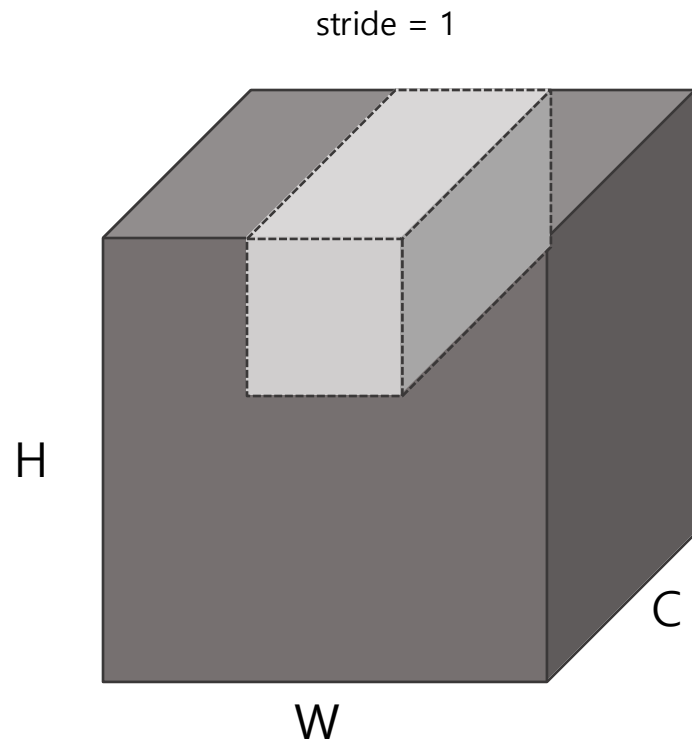
Convolution with Matrix Multiplication (called Convolution Lowering)



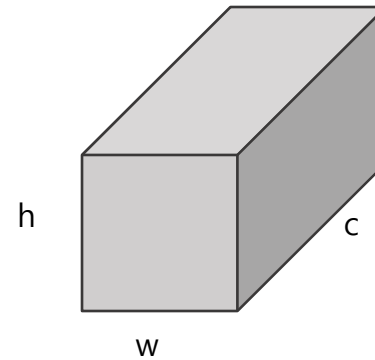
Convolution with Matrix Multiplication (called Convolution Lowering)



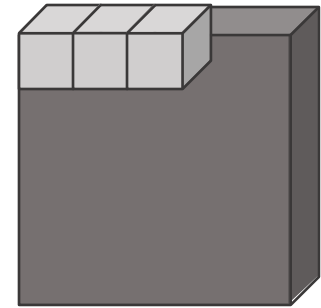
Convolution with Matrix Multiplication (called Convolution Lowering)



$*$



$=$



일반적인 컨볼루션 연산 형태

컨볼루션 연산은 행렬 곱 연산으로 변환될 수 있음

Convolution with Matrix Multiplication (called Convolution Lowering)

Input = 2 x 3 x 3

Filter = 1 x 2 x 2 x 2

Output = 1 x 2 x 2

Stride = 1

1	2	1
2	4	2
1	2	1

2	4	2
4	8	2
2	4	2

*

0.0	0.5
0.5	1.0

0	1
1	2

=

30.0

필터의 수가 1개인 예시

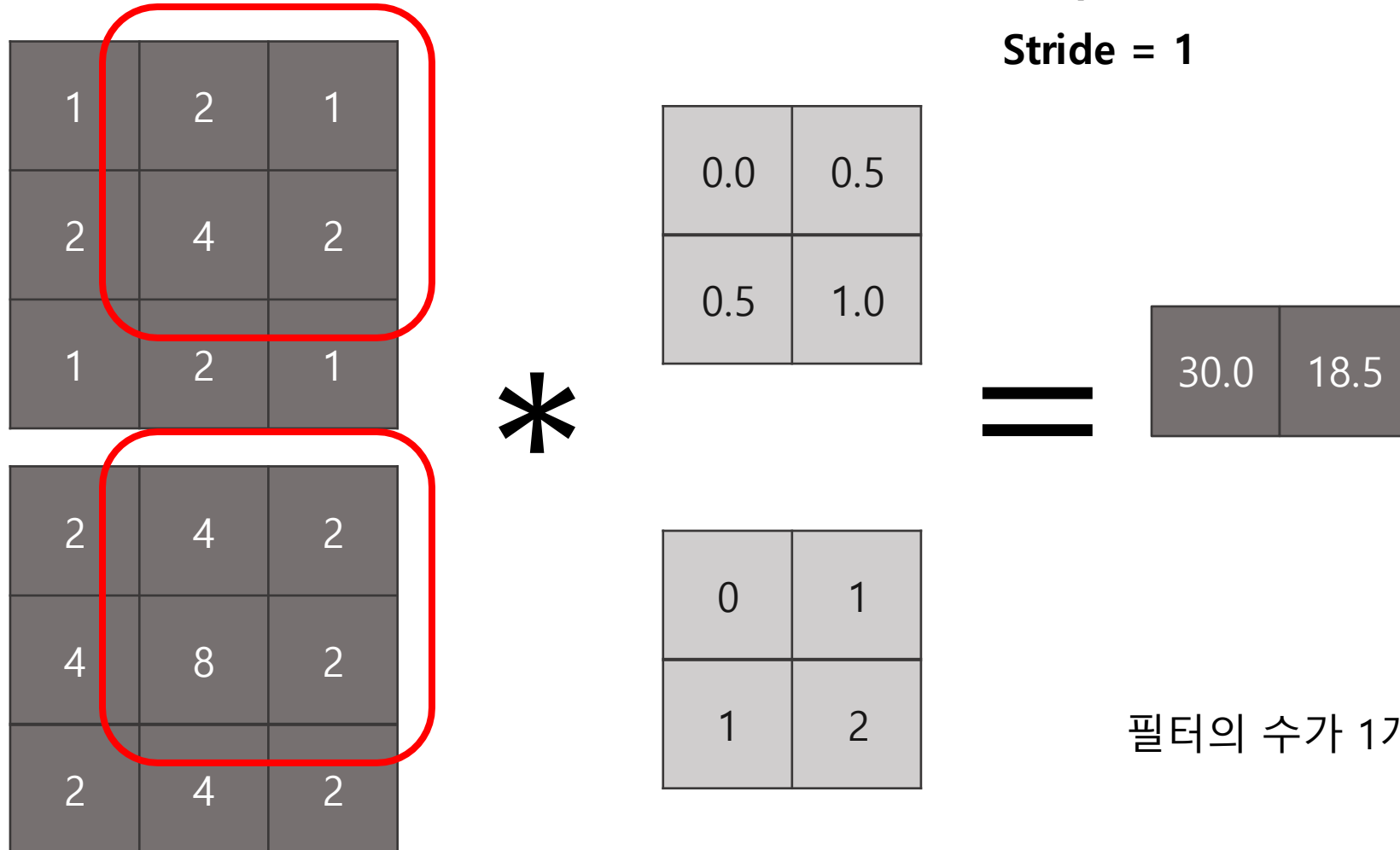
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = 2 x 3 x 3

Filter = 1 x 2 x 2 x 2

Output = 1 x 2 x 2

Stride = 1



필터의 수가 1개인 예시

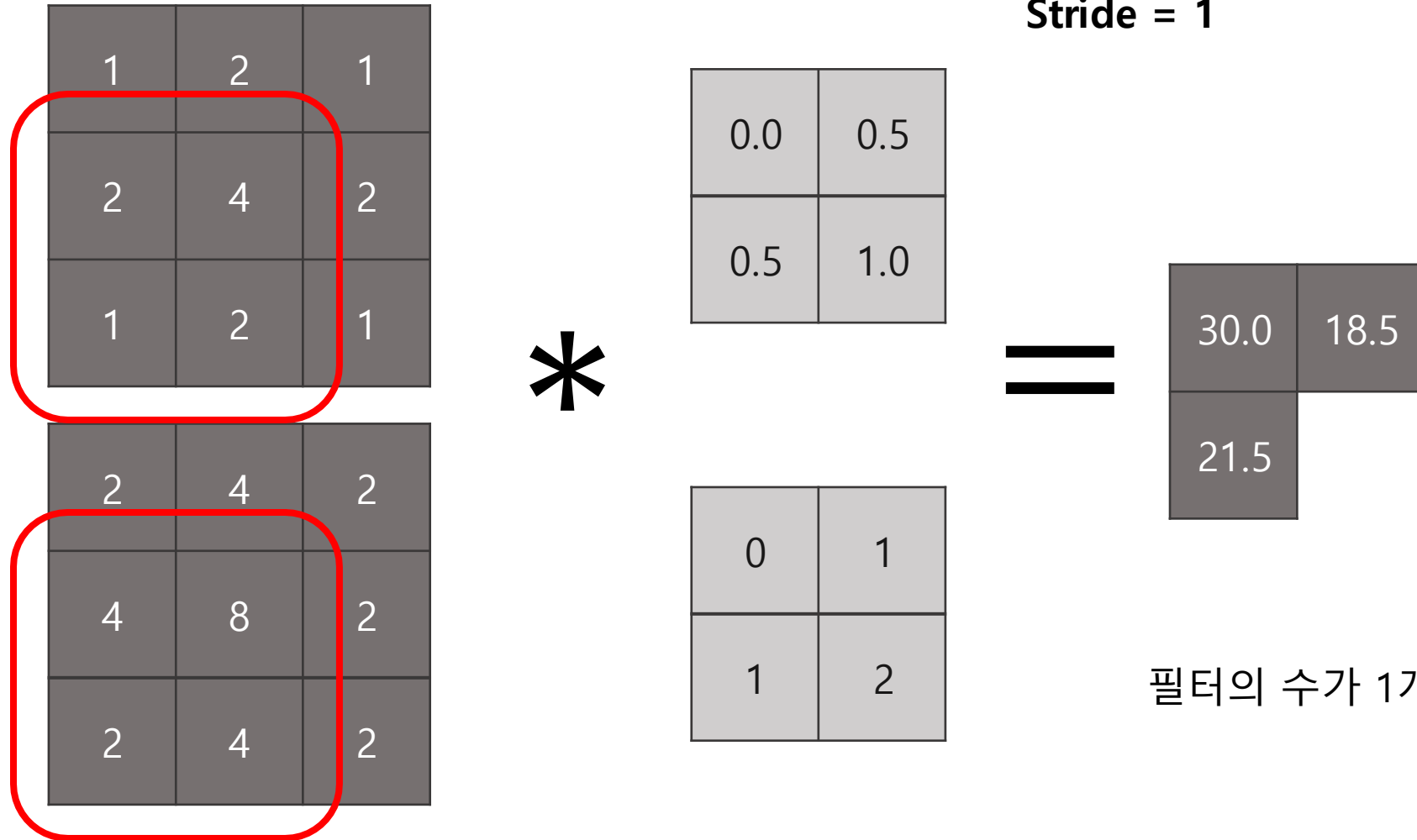
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = 2 x 3 x 3

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Stride = 1



필터의 수가 1개인 예시

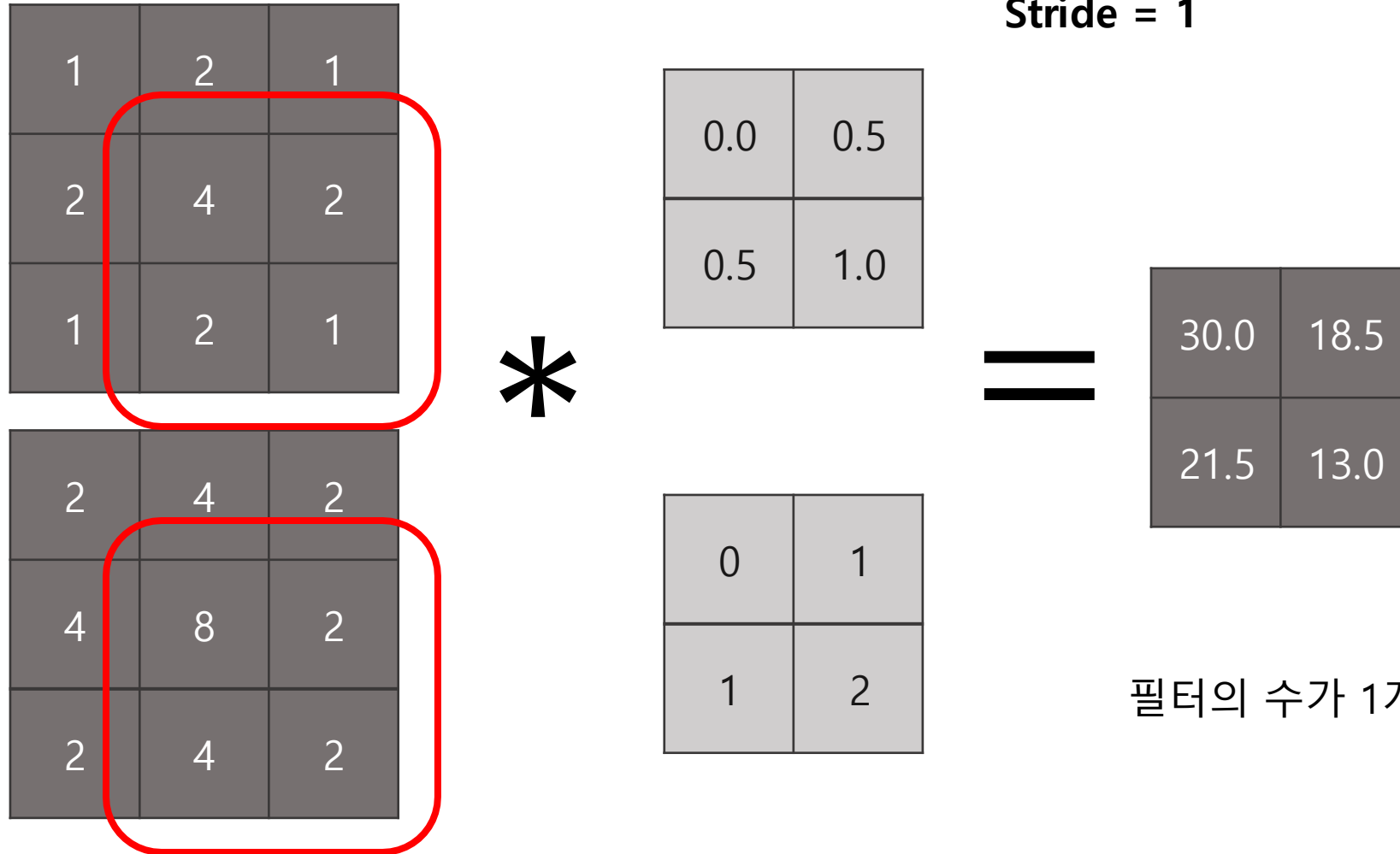
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = 2 x 3 x 3

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Output = 1 x 2 x 2

Stride = 1



필터의 수가 1개인 예시

Convolution with Matrix Multiplication (called Convolution Lowering)

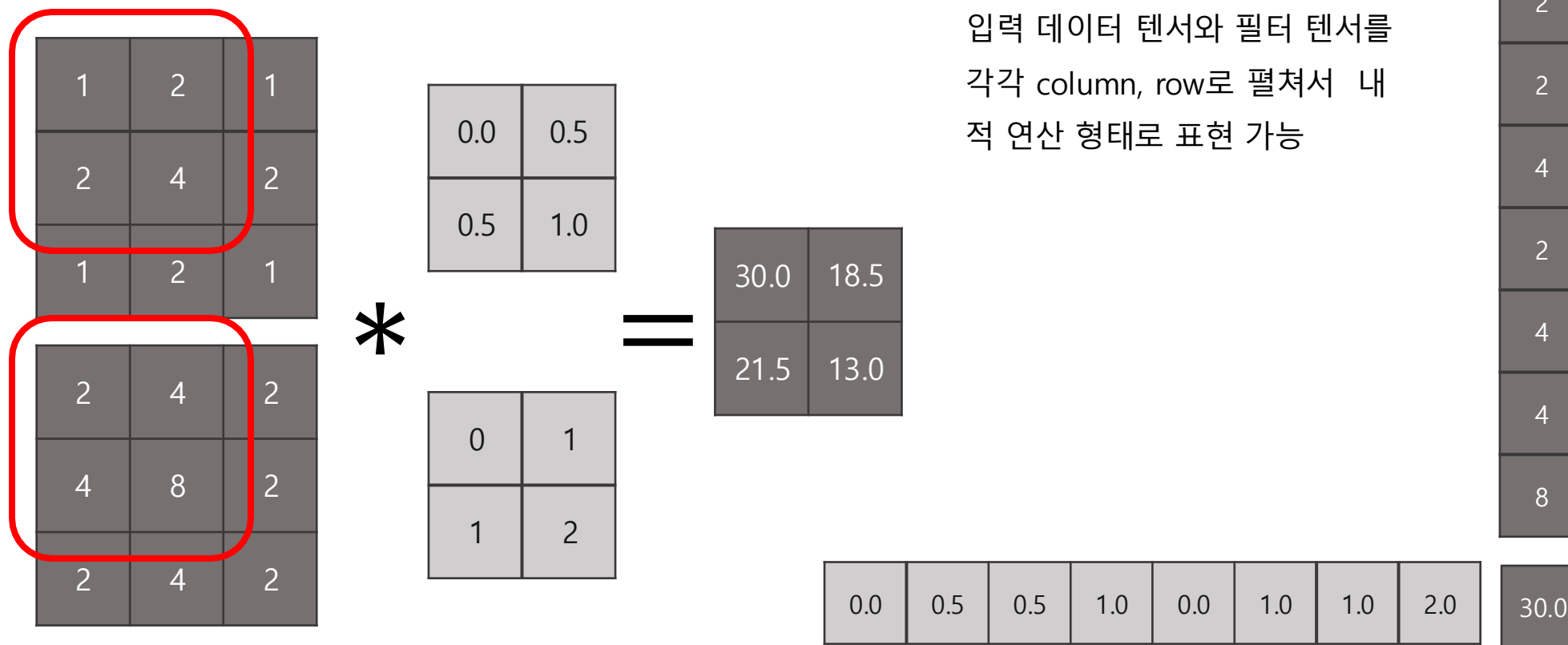
Input = $2 \times 3 \times 3$

Filter = $1 \times 2 \times 2 \times 2$

Output = $1 \times 2 \times 2$

Stride = 1

입력 데이터 텐서와 필터 텐서를
각각 column, row로 펼쳐서 내
적 연산 형태로 표현 가능



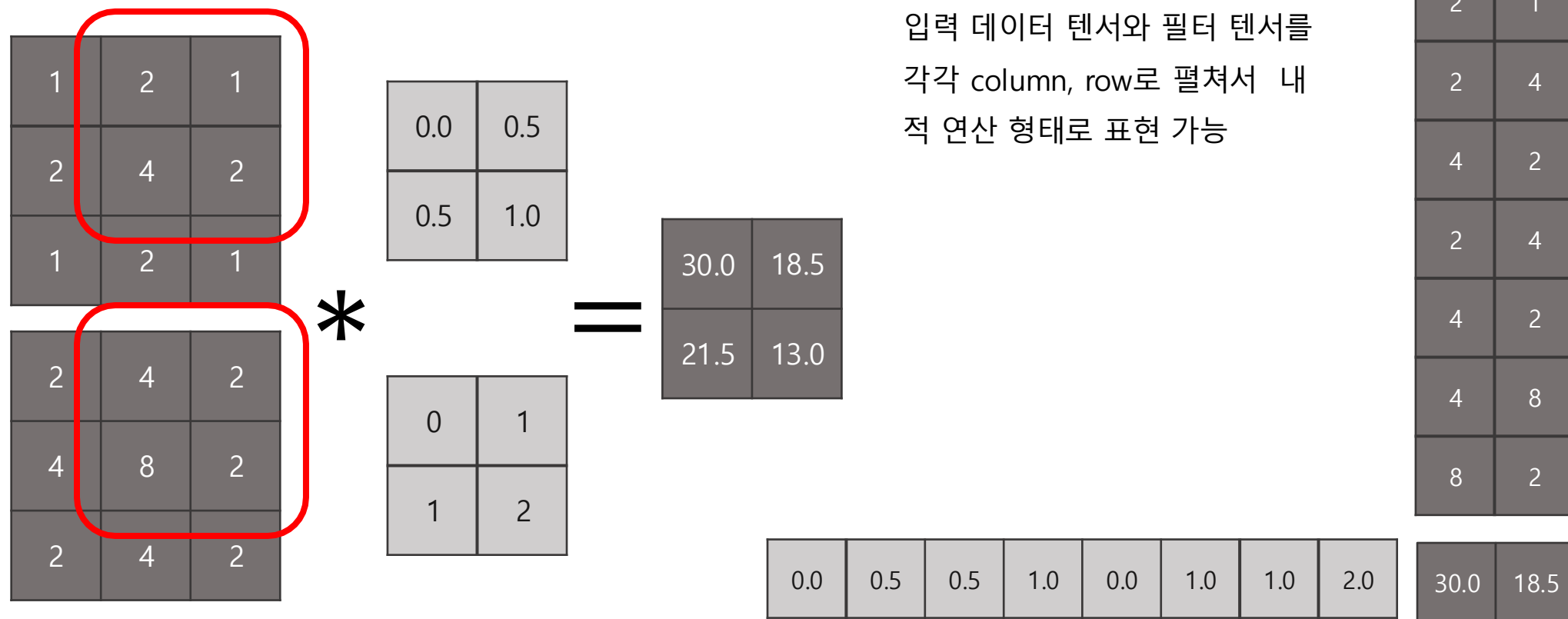
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = $2 \times 3 \times 3$

Filter = $1 \times 2 \times 2 \times 2$

Output = $1 \times 2 \times 2$

Stride = 1



Convolution with Matrix Multiplication (called Convolution Lowering)

Input = $2 \times 3 \times 3$

Filter = $1 \times 2 \times 2 \times 2$

Output = $1 \times 2 \times 2$

Stride = 1

입력 데이터 텐서와 필터 텐서를
각각 column, row로 펼쳐서 내
적 연산 형태로 표현 가능

1	2	1
2	4	2
1	2	1

2	4	2
4	8	2
2	4	2

*

0.0	0.5
0.5	1.0

0	1
1	2

=

30.0	18.5
21.5	13.0

0.0	0.5	0.5	1.0	0.0	1.0	1.0	2.0
-----	-----	-----	-----	-----	-----	-----	-----

1	2	2
2	1	4
2	4	1
4	2	2
2	4	4
4	2	8
4	8	2
8	2	4

30.0	18.5	21.5
------	------	------

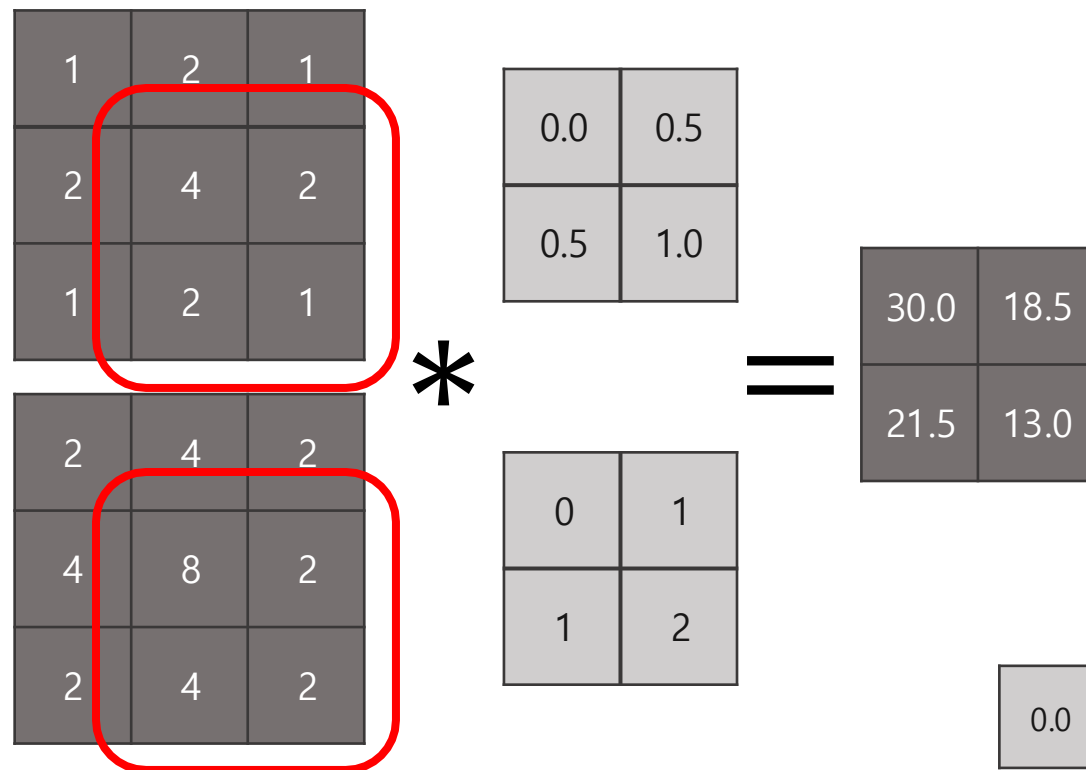
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = $2 \times 3 \times 3$

Filter = $1 \times 2 \times 2 \times 2$

Output = $1 \times 2 \times 2$

Stride = 1



입력 데이터 텐서와 필터 텐서를
각각 column, row로 펼쳐서 내
적 연산 형태로 표현 가능

0.0	0.5	0.5	1.0	0.0	1.0	1.0	2.0
-----	-----	-----	-----	-----	-----	-----	-----

1	2	2	4
2	1	4	2
2	4	1	2
4	2	2	1
2	4	4	8
4	2	8	2
4	8	2	4
8	2	4	2

30.0	18.5	21.5	13.0
------	------	------	------

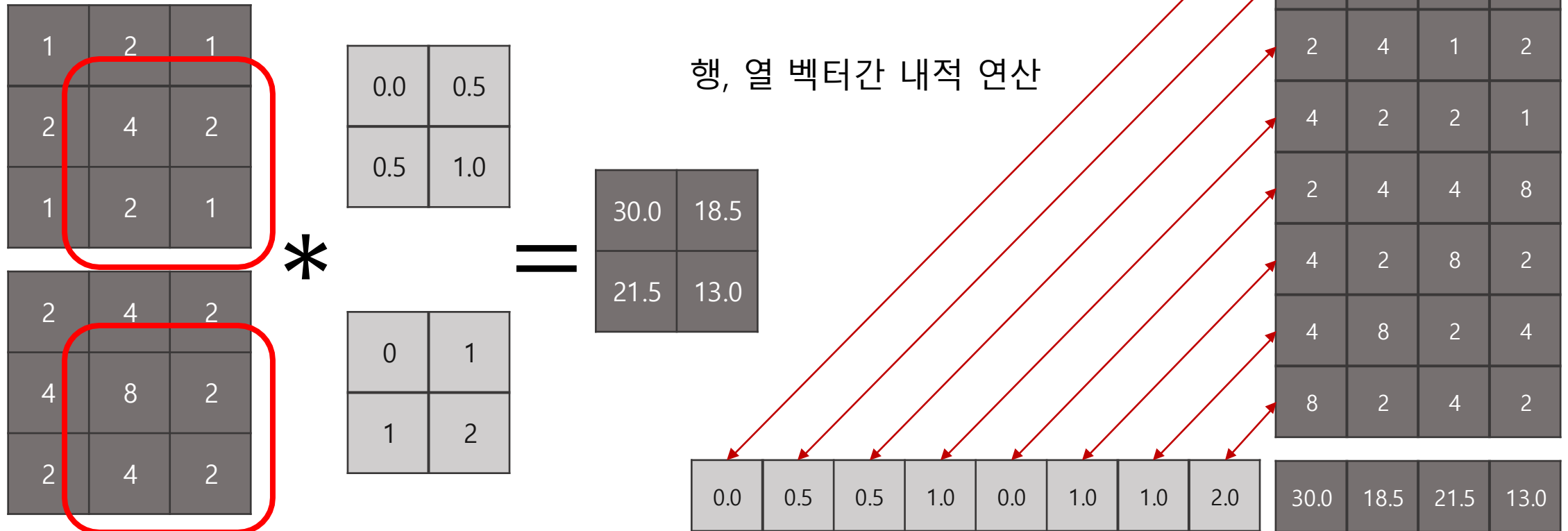
Convolution with Matrix Multiplication (called Convolution Lowering)

Input = $2 \times 3 \times 3$

Filter = $1 \times 2 \times 2 \times 2$

Output = $1 \times 2 \times 2$

Stride = 1



Convolution with Matrix Multiplication (called Convolution Lowering)

Input = 2 x 3 x 3

Filter = 1 x 2 x 2 x 2

Output = 1 x 2 x 2

Stride = 1



Input = 2 x 3 x 3

Filter = **N** x 2 x 2 x 2

Output = **N** x 2 x 2

Stride = 1

필터 1개에서 N개로 확장
= 입력과 필터의 행렬 연산

0.0	0.5	0.5	1.0	0.0	1.0	1.0	2.0

0.0	0.5	0.5	1.0	...	0.0	1.0	1.0	2.0
-----	-----	-----	-----	-----	-----	-----	-----	-----

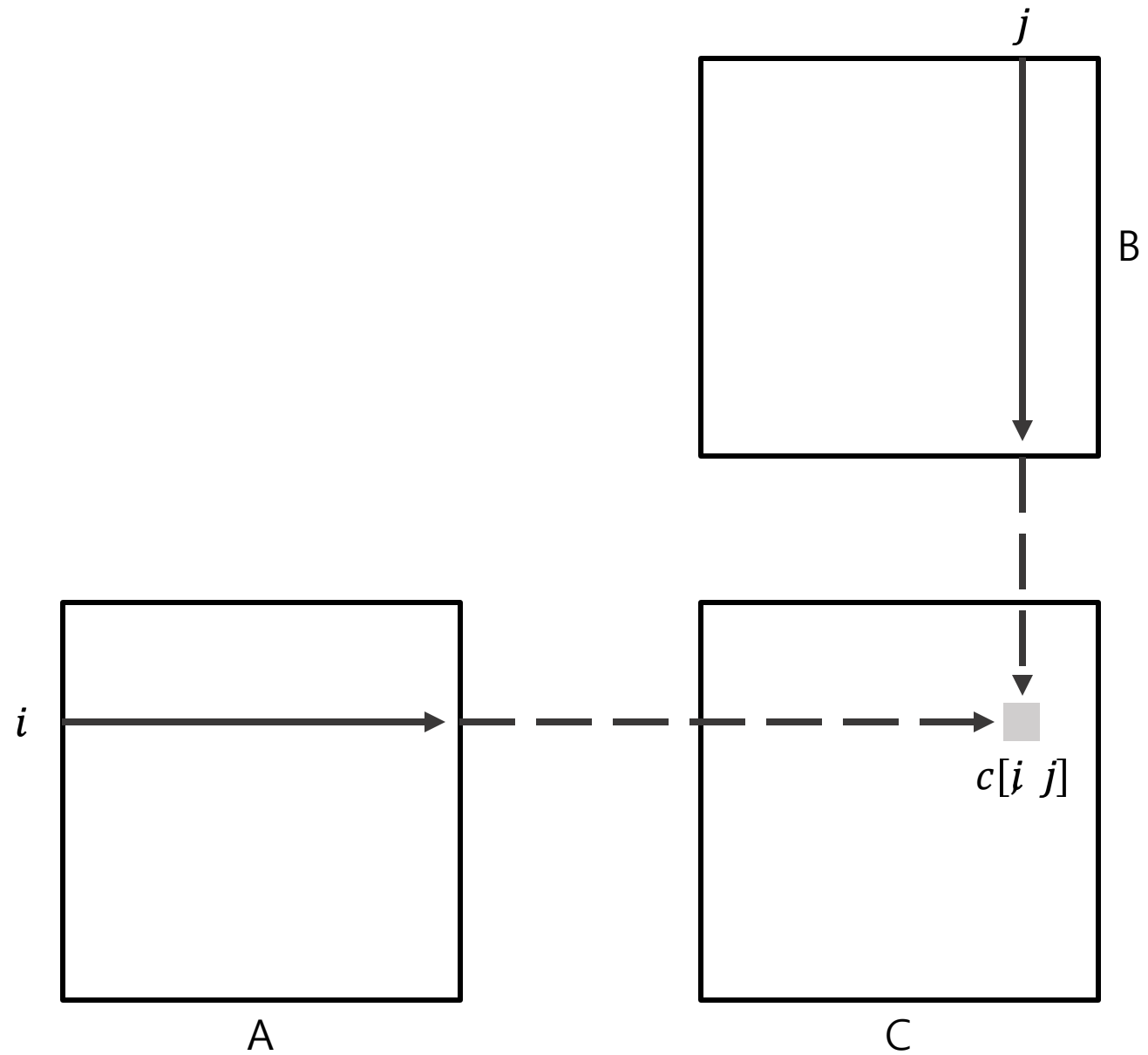
1	2	2	4
2	1	4	2
2	4	1	2
4	2	2	1
2	4	4	8
4	2	8	2
4	8	2	4
8	2	4	2

30.0	18.5	21.5	13.0

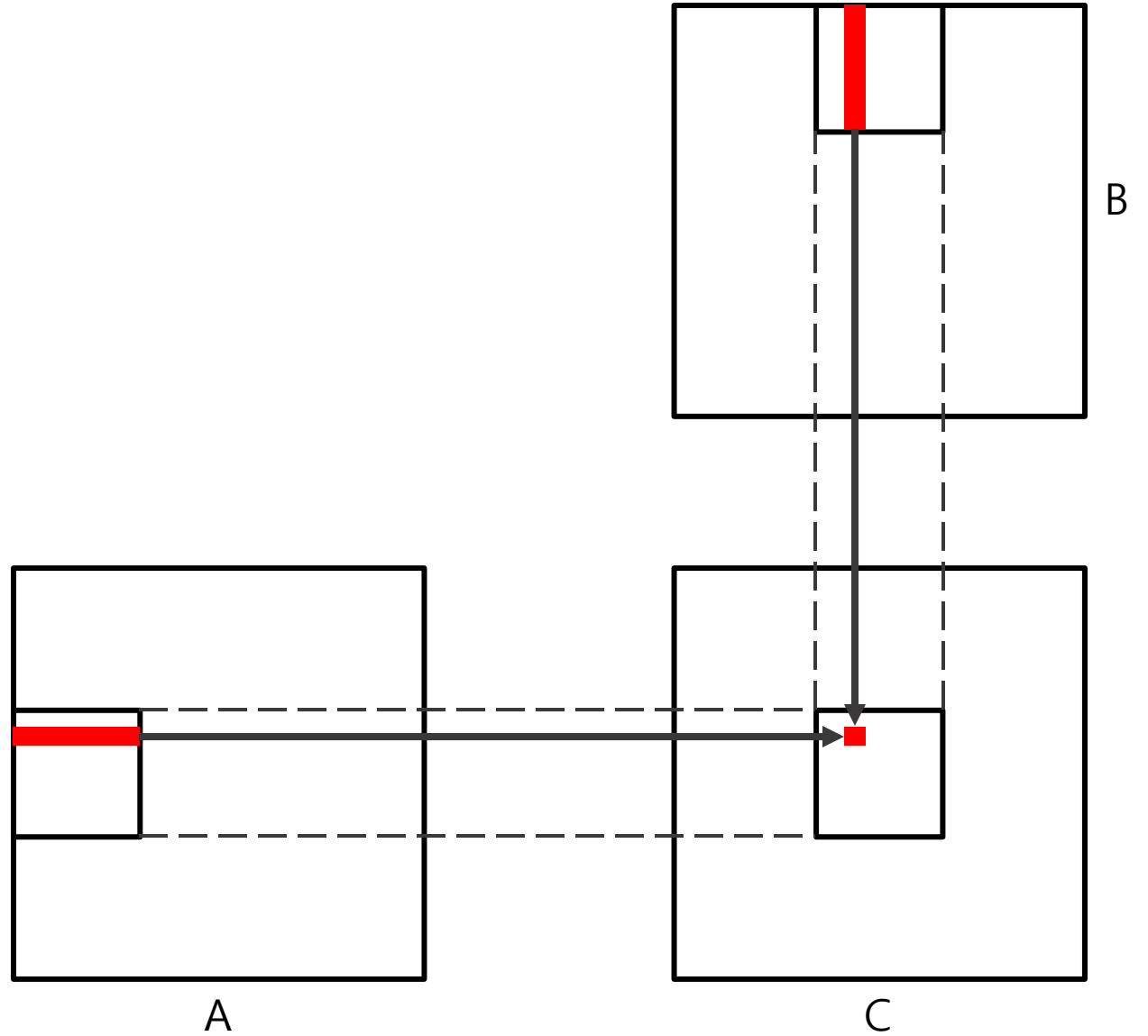
...

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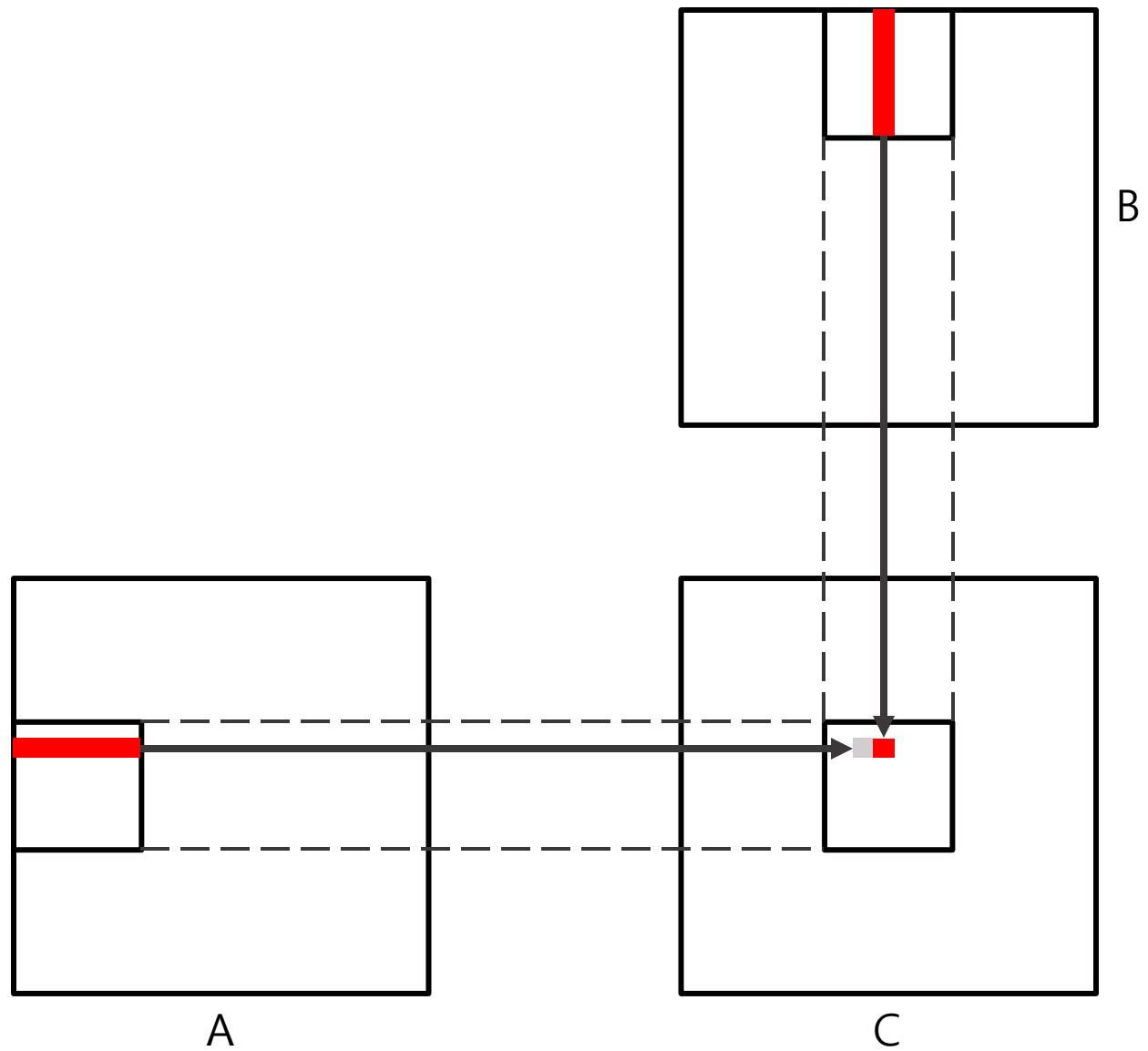
Matrix Tiling



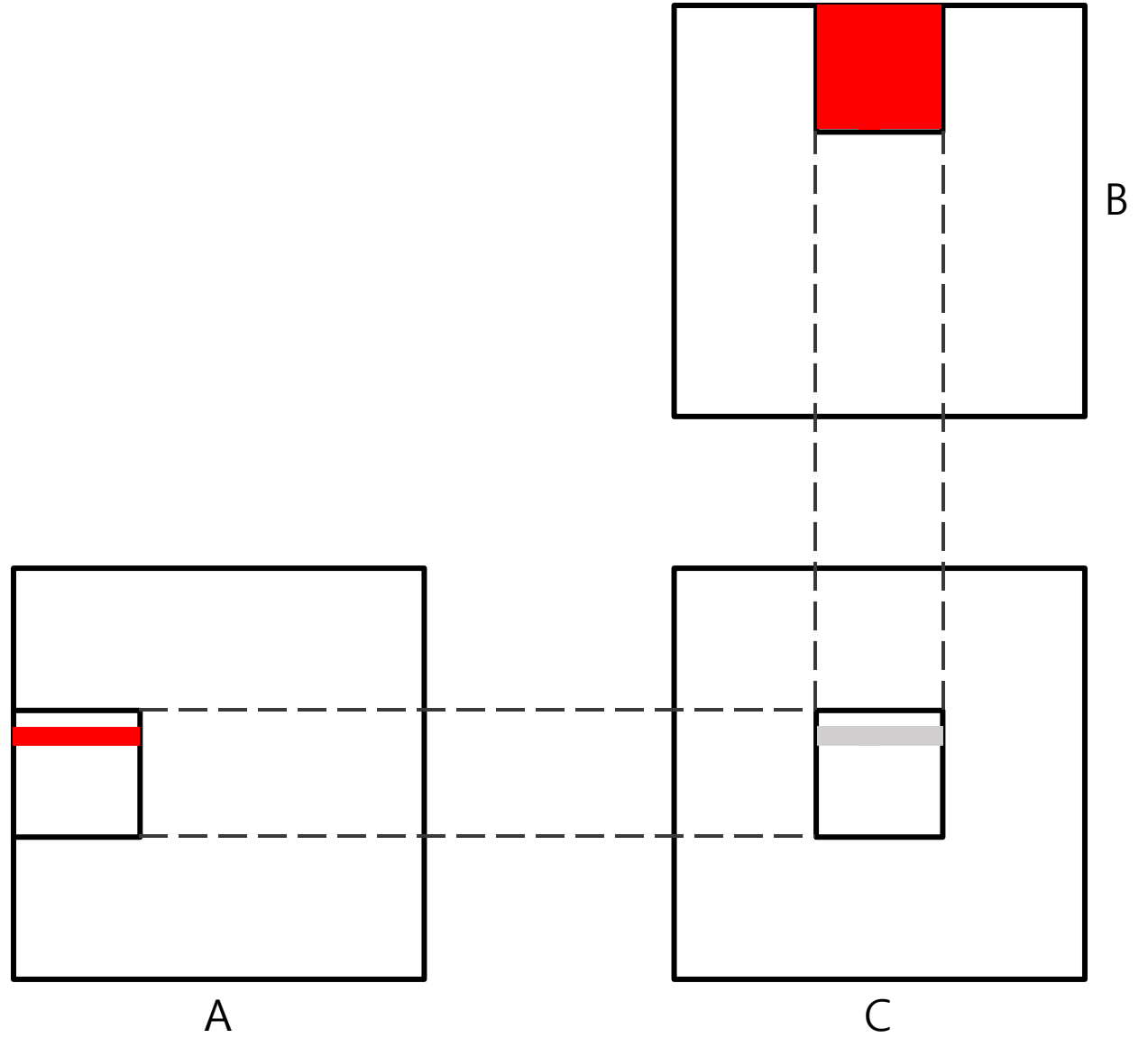
Matrix Tiling



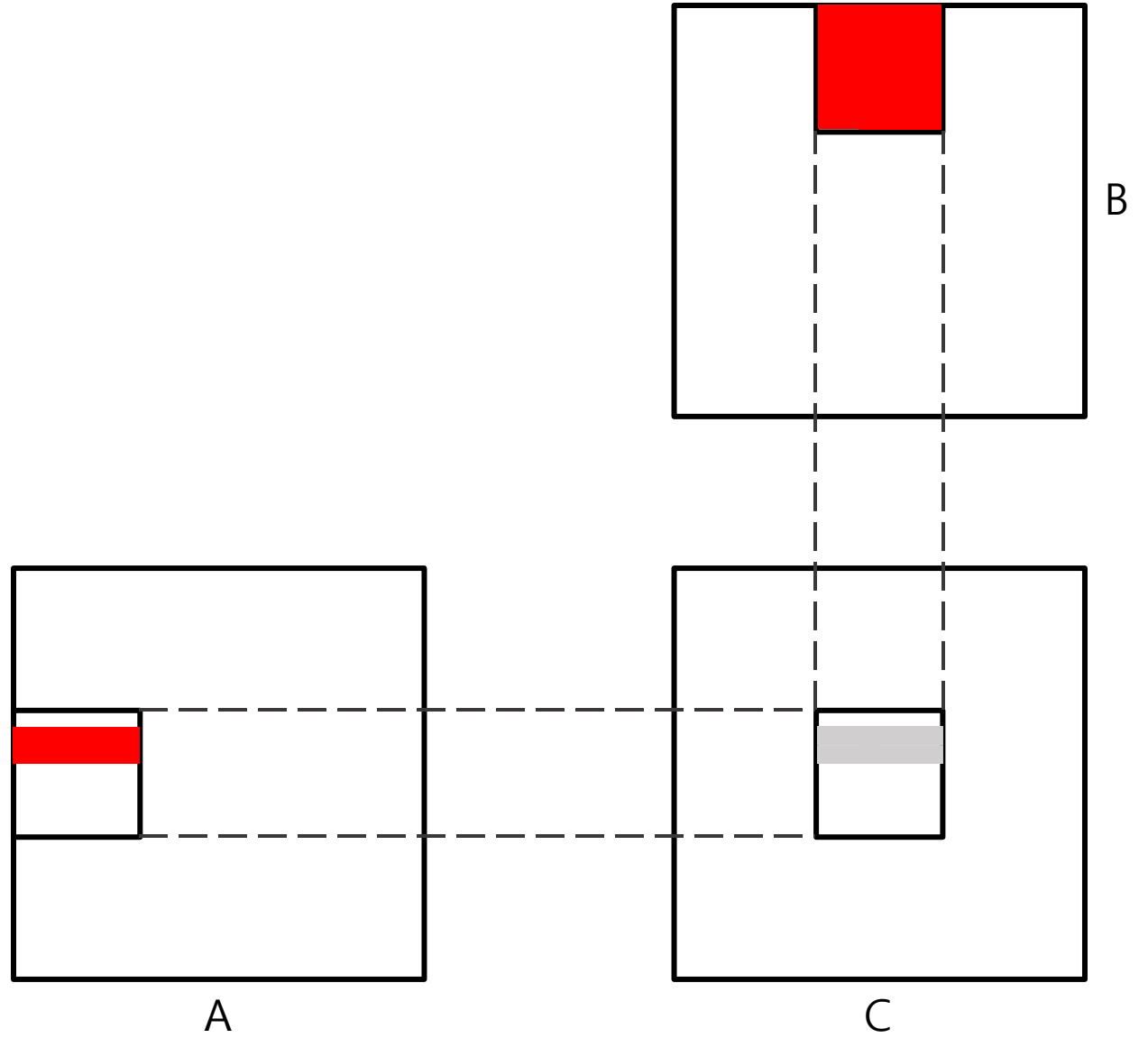
Matrix Tiling



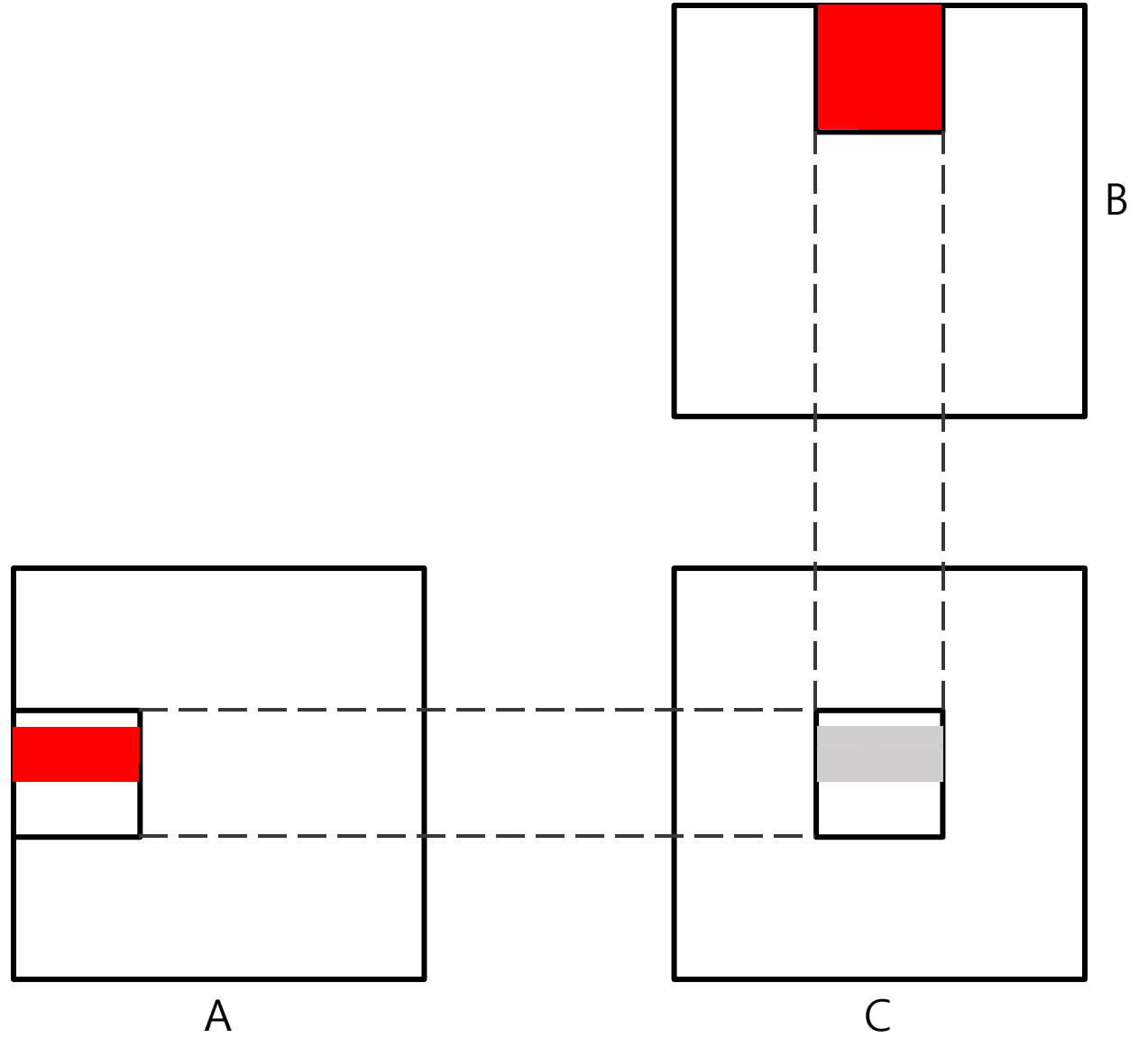
Matrix Tiling



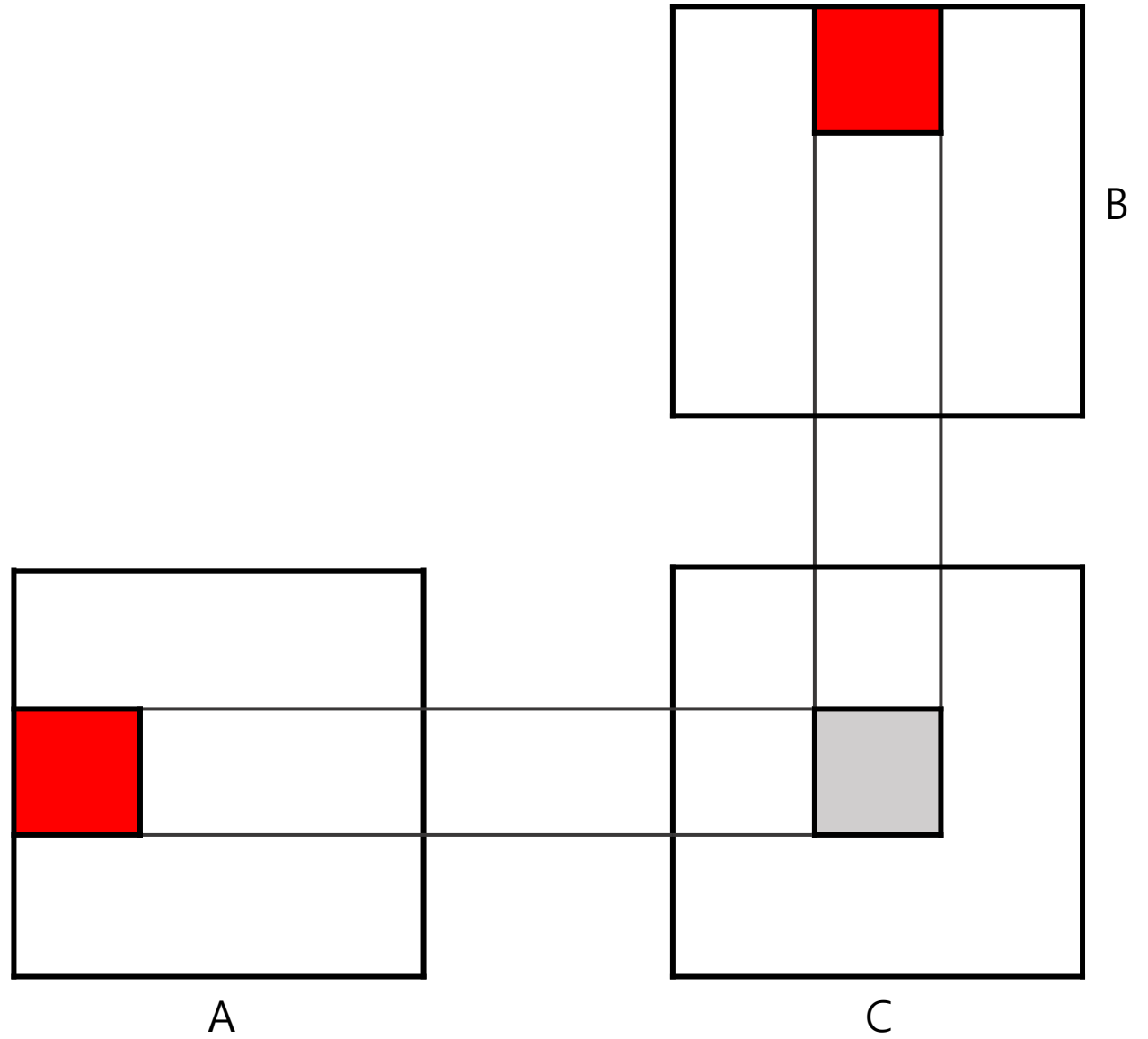
Matrix Tiling



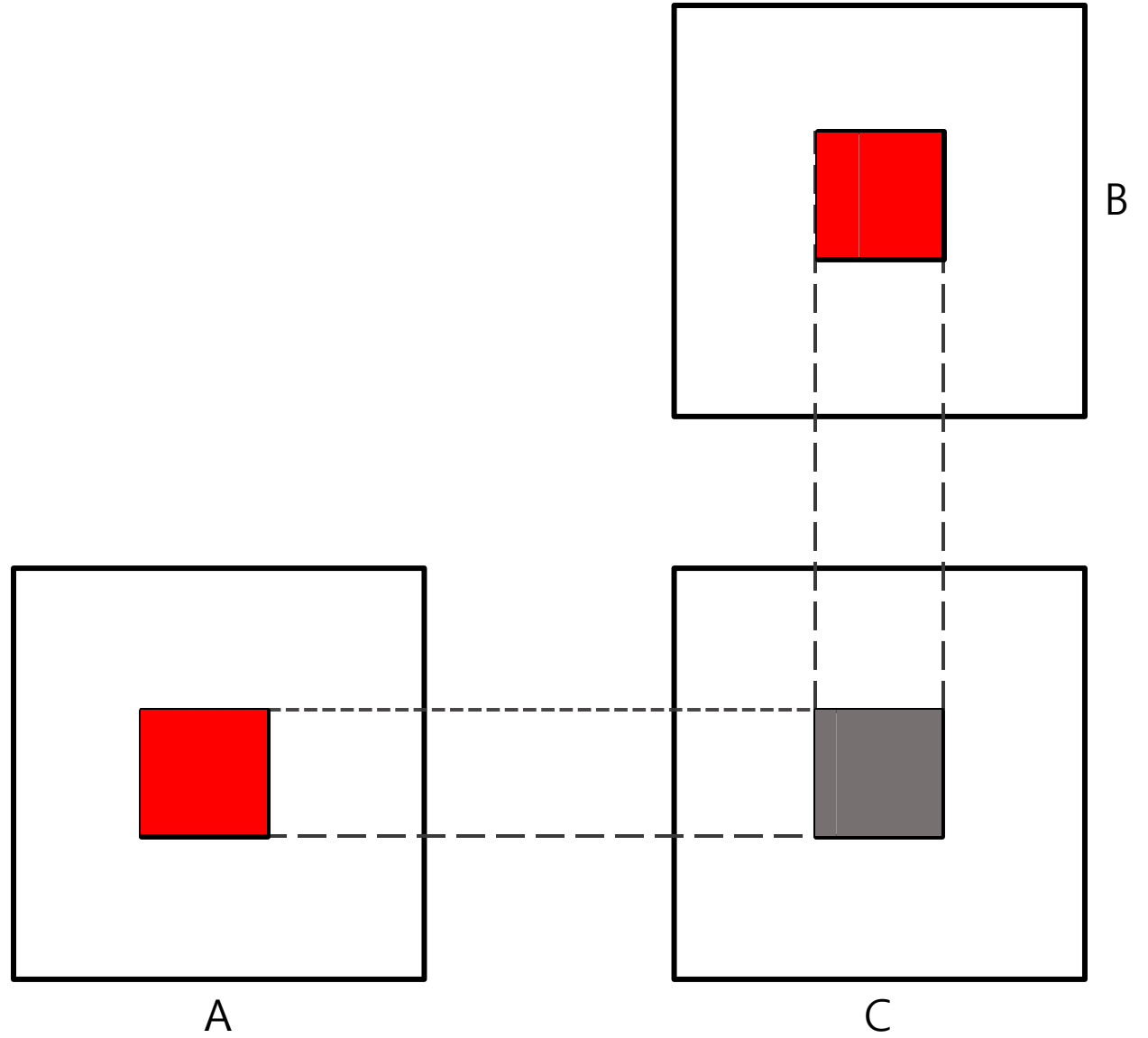
Matrix Tiling



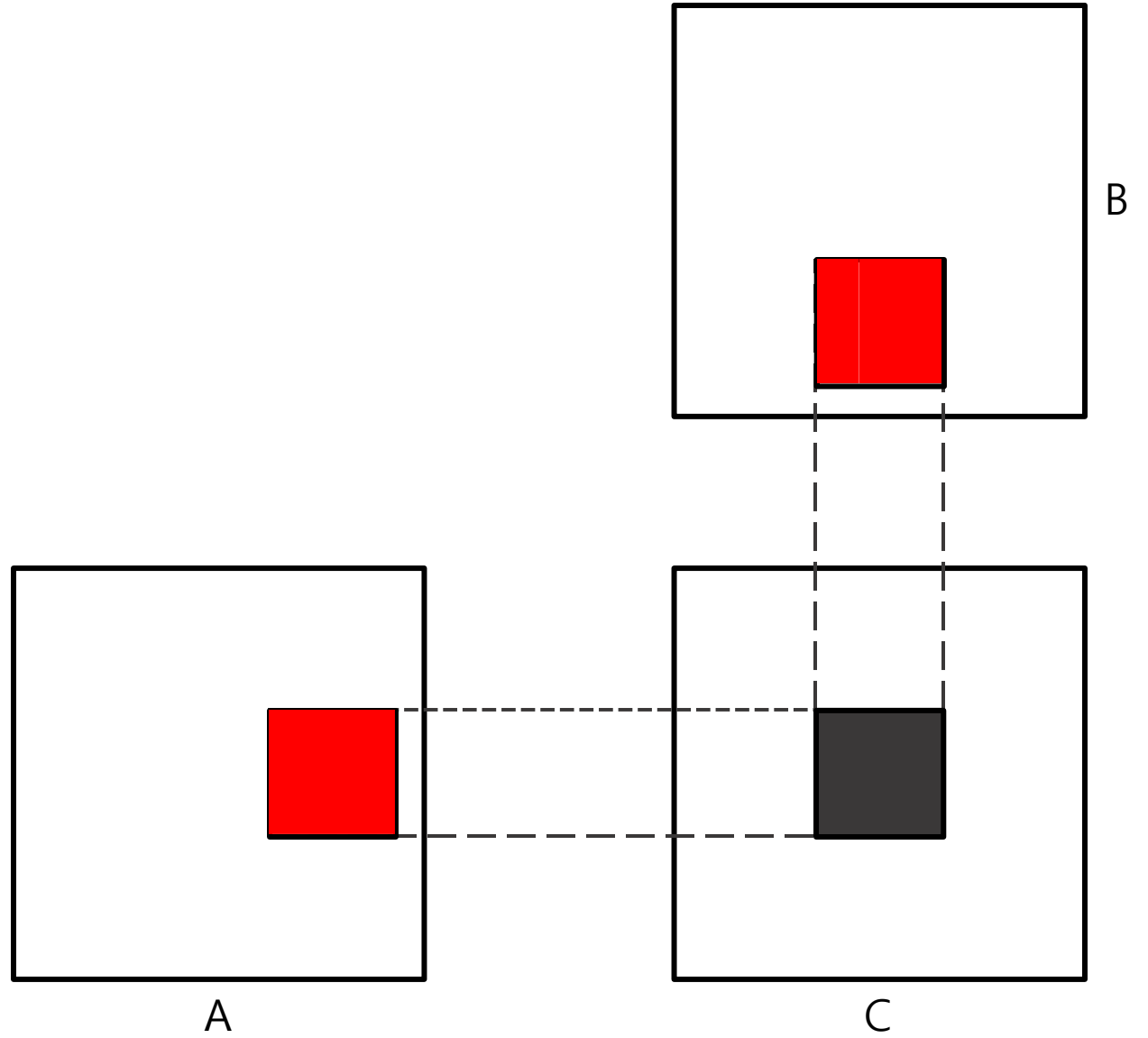
Matrix Tiling



Matrix Tiling



Matrix Tiling



Report

- **Requirements**
 - Briefly explain how your codes are implemented.
 - Briefly analyze your results.
 - Compare performance improvement of each step
 - Max 2 + 2 pages (without title)
 - Your own practice2.ipynb + document file (.pdf)
 - Report file name
 - {student_number}-{name}-p2.pdf, e.g, 2019-12345-홍길동-p1.pdf
 - Submit your report on eCampus
- **Due: Oct. 18th 23:59 KST**